

Biocorrosion of Ti-6Zr & Ti-8V Under Mechanically Disrupted Oxide Conditions

A comparative electrochemical assessment in simulated body fluid, referenced against the de-passivated (oxide-broken) wear scenario of Ti-6Al-4V.

TEST SCENARIO	Native TiO ₂ passive film mechanically fractured/removed by sliding, fretting, or third-body wear; bare alloy substrate exposed directly to SBF
REFERENCE BASELINE	Ti-6Al-4V (Grade 5) under tribocorrosion / fretting-corrosion conditions
ENVIRONMENT	Simulated Body Fluid (SBF), Hank's Balanced Salt Solution (HBSS), Phosphate Buffered Saline (PBS) — 37 °C, pH 7.4
METHOD BASIS	Open-circuit potential transients, potentiodynamic/potentiostatic polarisation, depassivation–repassivation current analysis, ion-release literature

ABSTRACT

This supplement addresses a specific and clinically important failure scenario: what happens electrochemically when the protective TiO₂ passive oxide is mechanically broken — by fretting, sliding wear, or third-body abrasion at an implant interface — and the bare metallic substrate is suddenly exposed to simulated body fluid (SBF). This is the **tribocorrosion** or **depassivation–repassivation** regime, distinct from static immersion corrosion, and it is the regime that governs real failure modes at modular implant junctions, fretting contacts, and articulating surfaces. **Ti-6Al-4V** is used as the reference baseline because it is the most extensively characterised alloy under exactly this condition. **Ti-6Zr** and **Ti-8V** are then

evaluated against this baseline using direct literature data where it exists for the named compositions, and using closely related compositional analogues (Ti-13Zr, Ti-15Zr, Ti-30Zr-5Nb-14Mo for the Zr system; Ti-12Mo-6Zr-2Fe "TMZF", Ti-3Fe, Ti-6Al-1V-3Fe for the V/Fe β -system) where direct data is unavailable — clearly flagged as such throughout. All quantitative claims are cited inline.

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SECTION 1

The Depassivation–Repassivation Paradigm

All commercial titanium alloys owe their corrosion resistance to a thin (2–10 nm), spontaneously formed, self-healing oxide film — predominantly TiO₂ — that forms within milliseconds on exposure to air or aqueous media.^[1] Static immersion corrosion testing (simple electrode exposed to SBF, no mechanical contact) characterises this *intact* oxide and is the basis of the property data presented in the main report. However, in service, implant components are frequently subjected to micromotion, fretting, or sliding contact at modular junctions (e.g., femoral head–neck tapers, screw–plate interfaces, articulating bearing surfaces) that mechanically fracture and remove the oxide film locally, exposing bare metal directly to the surrounding biological fluid.^[2,3]

This combined mechanical–electrochemical degradation mode is termed **tribocorrosion** (or, in the specific case of small-amplitude oscillatory contact, **fretting corrosion**).

sion).^[1,2] It is fundamentally different from static corrosion in three respects relevant to this comparison:

01	Depassivation: Mechanical contact strips the oxide within the contact zone, transiently exposing base-metal atoms with no protective barrier. Measured current density during active sliding typically rises ~100× above the steady-state passive current. ^[4]
02	Open-circuit potential (OCP) collapse: The corrosion potential drops sharply and cathodically (ΔE) the instant the oxide is breached — the magnitude of this drop is a direct, widely used index of how vulnerable an alloy's passive film is to mechanical disruption. ^[5]
03	Repassivation: Once sliding contact moves on or stops, the bare metal re-oxidises and the OCP recovers toward (but rarely fully to) its pre-sliding value. The rate and completeness of this recovery — repassivation kinetics — determines the cumulative ion dose released over a duty cycle of repeated micromotion. ^[5,6]

It is this disrupted-oxide condition — not the intact static-immersion condition — that the present comparison addresses, per the scenario specified: the TiO₂ layer is assumed mechanically broken, and the bare alloy substrate is exposed directly to SBF.

SECTION 2

Reference Baseline — Ti-6Al-4V With Mechanically Broken Oxide in SBF

2.1 Passive Film Composition Prior to Disruption

The native passive film on Ti-6Al-4V is a duplex structure: an outer porous layer and inner barrier layer, composed primarily of TiO₂ with co-existing **Al₂O₃** and **V₂O₅** incorporated from the alloy's substitutional elements.^[7,8] This is the critical chemical distinction from the two alloys under comparison: Al₂O₃ and V₂O₅ are both present in the Ti-6Al-4V passive film, and **both dissolve** under physiological and mechanically disturbed conditions, leaving the underlying film more defect-prone than a film built from more stable oxide formers.^[7]

2.2 Depassivation Behaviour Under Mechanical Disruption

TI-6AL-4V — TRIBOCORROSION / DEPASSIVATION PARAMETERS IN SBF		
STEADY-STATE PASSIVE CURRENT	$< 1 \times 10^{-6} \text{ A}$	<i>Stable baseline before mechanical contact begins [4]</i>
CURRENT DURING ACTIVE SLIDING	$\sim 100\times$ passive current	<i>Sharp ascent/descent on each contact pass; reflects rapid depassivation followed by fast repassivation [4]</i>
OCP DROP (ΔE) AT HIGH LOAD (17.5 N)	$658 \pm 64 \text{ mV}$	<i>Highest potential drop recorded among compared alloys in a multi-alloy SBF tribocorrosion study — indicates the least mechanically resilient passive film of the group tested [5]</i>
OCP RECOVERY (REPASSIVATION) POST-SLIDING	$+0.095 \text{ V}$ anodic	<i>Partial recovery toward pre-sliding value; does not fully restore baseline OCP within typical test windows [6]</i>
SPECIFIC WEAR RATE (NORMAL GLUCOSE HBSS)	$1.68 \times 10^{-6} \text{ mm}^3/\text{N}\cdot\text{m}$	<i>Lowest wear rate among an Fe-modified Ti alloy comparison set — Ti-6Al-4V's $\alpha+\beta$ duplex structure provides comparatively good strain-hardening response under sliding contact [6]</i>
DISSOLVED SPECIES IN SBF (BROKEN-OXIDE)	Ti^{4+} , Al^{3+} , V^{5+} (vanadate)	<i>Al_2O_3 and V_2O_5 in the disrupted film dissolve preferentially, releasing Al^{3+} and vanadate ions into the surrounding fluid [7,9]</i>

2.3 Ion Release and Toxicological Context

Once the passive film is mechanically breached, the exposed $\text{Al}_2\text{O}_3/\text{V}_2\text{O}_5$ -bearing oxide and the underlying alloy release aluminium and vanadium species into the surrounding biological fluid. [9] Vanadium in its higher oxidation states (V^{4+} , V^{5+} , forming vanadate, VO_4^{3-}) is the more biologically active and soluble species; chronic exposure to released Al^{3+} and vanadate ions has been associated in the clinical literature with peripheral neuropathy, osteomalacia, and a proposed (though not conclusively established) association with Alzheimer's-type neurological effects. [9,10] Critically for the tribocorrosion scenario specifically: because mechanical disruption continuously regenerates fresh, unprotected metal surface at the contact zone, the *cumulative* ion dose

under repeated fretting is substantially higher than under static immersion, even though the per-event dissolution rate at any single instant is comparable. ^[2,9]

BASELINE SUMMARY — Ti-6Al-4V

Under the disrupted-oxide scenario, Ti-6Al-4V exhibits the largest measured OCP collapse among directly compared titanium implant alloys ^[5], releases two biologically active species (Al^{3+} and vanadate) rather than one, and — despite comparatively good wear resistance due to $\alpha+\beta$ strain hardening — its passive film chemistry is structurally the weakest link of the three alloys assessed here. This is the baseline against which Ti-6Zr and Ti-8V are now compared.

SECTION 3

Ti-6Zr — Biocorrosion Under the Same Disrupted-Oxide Scenario

No published tribocorrosion study isolates the binary Ti-6Zr composition specifically under mechanical oxide disruption in SBF. The closest direct analogues in the literature are Ti-13Zr, Ti-15Zr, Ti-15Zr-5Mo, and the more heavily alloyed Ti-30Zr-5Nb-14Mo — all evaluated head-to-head against Ti-6Al-4V under identical sliding/fretting protocols in SBF or PBS. ^[3,11,12] Because Zr is isomorphous with Ti and forms a chemically similar, even more stable oxide, the trends from these Zr-bearing alloys are directly extrapolable to the lower-Zr Ti-6Zr composition; the magnitude of the benefit would be expected to scale down somewhat at 6 wt% Zr relative to the 13–30 wt% Zr alloys cited, but the *direction* of every effect described below is consistent and mechanistically well-founded.

3.1 Passive Film Composition — The Decisive Chemical Difference

Where Ti-6Al-4V's broken oxide contains Al_2O_3 and V_2O_5 (both prone to dissolution), Ti-6Zr's broken oxide contains only **TiO₂** and **ZrO₂**. Comparative thermodynamic and electrochemical assessment confirms that ZrO₂ — along with Nb₂O₅ and Ta₂O₅ in other β -stabilised systems — exhibits markedly better chemical stability and corrosion resistance than TiO₂ itself. ^[7] This means that when the Ti-6Zr oxide is me-

chanically fractured, the residual and re-forming film is chemically more robust at every stage of the depassivation–repassivation cycle, not just in the undisturbed state.

3.2 Depassivation and Repassivation Kinetics

TI-6ZR (EXTRAPOLATED FROM TI-ZR ALLOY FAMILY) – TRIBOCORROSION PARAMETERS		
OCP DROP (ΔE) UNDER SLIDING	Substantially smaller than Ti-6Al-4V	<i>The Zr-richer Ti-30Zr-5Nb-14Mo showed ΔE of 345–416 mV vs. 658 mV for Ti-6Al-4V at equivalent loads — roughly half the potential collapse [5]</i>
REPASSIVATION CAPACITY	Superior	<i>Ti-15Zr and Ti-15Zr-5Mo demonstrated "a better recovery capacity of the passive oxide layer" relative to CP-Ti Grade 4 in directly comparable fretting-corrosion testing [3]</i>
VOLUME / MATERIAL LOSS IN SBF TRIBOCORROSION	Favourable	<i>β-stabilised Zr-Nb-bearing alloy (Ti-13Nb-13Zr) recorded lower volume loss and lower friction coefficient than the comparator set including Ti-6Al-4V in matched SBF tribocorrosion testing [13]</i>
DISSOLVED SPECIES IN SBF (BROKEN-OXIDE)	Ti⁴⁺, Zr⁴⁺ only	<i>Both are classed as biophile, essentially non-toxic species at physiological concentrations — no analogue to the Al³⁺/vanadate release seen with Ti-6Al-4V [14]</i>
CYTOTOXICITY OF RELEASED SPECIES	None observed	<i>In-vitro cell viability for Ti-Zr alloy extraction media consistently exceeds 90% across compositions; no neurotoxic ion pathway exists for Zr as it does for V [14,15]</i>

3.3 Hardness, Wear Resistance, and the Threshold for Oxide Breakage

An additional, second-order consideration is that Ti-Zr alloys are measurably harder than cp-Ti and comparable to or modestly below Ti-6Al-4V (240–280 HV for Ti-6Zr vs. ~330–350 HV for Ti-6Al-4V), meaning the mechanical threshold force required to initiate oxide fracture in the first place may be somewhat lower than for Ti-6Al-4V.

However, this is outweighed in clinical relevance by the fact that *once* the oxide is broken, the chemistry of what is released, and the speed at which protection is restored, both favour Ti-6Zr decisively. ^[3,7]

TI-6ZR UNDER DISRUPTED-OXIDE SCENARIO — SUMMARY

Relative to the Ti-6Al-4V baseline, the Ti-Zr family shows a smaller OCP collapse on depassivation, faster and more complete repassivation, lower cumulative material loss in matched SBF tribocorrosion tests, and — most importantly from a biocorrosion standpoint — releases no biologically active or neurotoxic ionic species when the oxide is breached. This makes Ti-6Zr the more favourable choice of the two alloys under exactly the mechanically disrupted scenario specified.

SECTION 4

Ti-8V / Ti185 — Biocorrosion Under the Same Disrupted-Oxide Scenario

As with Ti-6Zr, no published tribocorrosion study isolates the Ti-8V binary or Ti185 (Ti-1Al-8V-5Fe) composition specifically under mechanical oxide disruption in SBF — Ti185's literature base is concentrated on aerospace mechanical properties rather than biomedical electrochemistry, reflecting its primary industrial application (Section 7 of the main report). The most relevant analogues for this comparison are other Fe- and V-bearing metastable β -titanium alloys evaluated directly against Ti-6Al-4V under wear-corrosion conditions in SBF/HBSS: **TMZF (Ti-12Mo-6Zr-2Fe)**, **Ti-3Fe**, and **Ti-6Al-1V-3Fe**. ^[6,16] These provide the closest available evidence base for how a high- β -stability, Fe/V-bearing alloy behaves once its oxide is broken in a biological fluid — and the picture that emerges is considerably more cautionary than for the Zr system.

4.1 Passive Film Composition — A Compounding Concern

Ti185 contains 8 wt% V — double the vanadium content of Ti-6Al-4V (4 wt%) — plus 5 wt% Fe. The passive film forming on V-bearing β -alloys incorporates V_2O_5 in

proportion to bulk V content, and V₂O₅ is confirmed to dissolve readily, behaving as a Lux-Flood acidic oxide that is progressively displaced/attacked under aqueous and chloride-containing conditions.^[8,17] With double the vanadium of the baseline alloy, the disrupted oxide on Ti-8V/Ti185 would be expected — on direct compositional grounds — to release a correspondingly larger reservoir of dissolvable V₂O₅, all else equal. Iron additions complicate this further: published work confirms that for Ti-xFe systems, **passive current density increases with Fe content**, and Fe-bearing oxide films tend toward poorer corrosion resistance than the unmodified base alloy, though low-Fe substitutions for V (e.g., Ti-6Al-4Fe, Ti-5Al-2.5Fe) show comparable passivation behaviour to Ti-6Al-4V at lower Fe loadings.^[16]

4.2 Wear-Corrosion Synergy — The Critical β -Phase Vulnerability

The single most important and best-documented finding for this alloy class is a direct mechanistic study of **TMZF (Ti-12Mo-6Zr-2Fe)** — a metastable β -titanium alloy of the same general class as Ti185 — conducted specifically to understand an FDA recall driven by unacceptable in-service wear.^[18] The study's central finding has direct relevance here:

KEY FINDING — β -PHASE WEAR-CORROSION SYNERGY

TMZF's dry wear behaviour (oxide intact, no SBF) is statistically indistinguishable from Ti-6Al-4V's, consistent with their similar hardness. **But in SBF, with the oxide subject to repeated mechanical disruption, TMZF's wear is significantly accelerated relative to Ti-6Al-4V** — a divergence not predicted by hardness or dry-wear behaviour alone. The mechanism identified is the **lack of strain-hardening capacity characteristic of single-phase β -titanium microstructures**, in contrast to the $\alpha+\beta$ duplex structure of Ti-6Al-4V, which work-hardens at the contact zone and partially resists progressive material removal once the oxide is broken.^[18]

This finding is mechanistically transferable to Ti185/Ti-8V: both TMZF and Ti185 are predominantly single-phase, metastable BCC β microstructures in their service condition (Section 5 of the main report), both rely on substitutional β -stabilisers (Mo/Fe for TMZF; V/Fe for Ti185) rather than a duplex $\alpha+\beta$ structure, and neither possesses the intrinsic work-hardening response of Ti-6Al-4V's $\alpha+\beta$ matrix. The precipitation-hardened condition of Ti185 (Section 4.4 of the main report) increases bulk hardness substantially, which may partially offset this vulnerability — but the un-

derlying single-phase- β strain-hardening deficit identified in TMZF has not been shown to be eliminated by aging treatment, and no direct SBF tribocorrosion data exists to confirm whether STA-aged Ti185 escapes this effect.

4.3 Depassivation Behaviour of Related Fe-Bearing Alloys

FE/V-BEARING B-TI ANALOGUES – TRIBOCORROSION PARAMETERS IN SBF/HBSS		
PEAK SLIDING CURRENT DENSITY	Ti-3Fe: 6.4×10^{-4} A	<i>Highest of an Fe-modified Ti comparison set; Ti-6Al-1V-3Fe and Ti-6Al-4V remained below 4.3×10^{-4} A — higher Fe content alone correlates with weaker depassivation resistance [6]</i>
OCP DROP ON SLIDING (TI-3FE)	0.61 V (-0.42→-1.03 V)	<i>Larger collapse than the Ti-6Al-4V baseline (which itself showed the largest ΔE of the Zr-bearing comparator set in Section 3) [6]</i>
OCP RECOVERY (REPASSIVATION, TI-3FE)	< 0.01 V — negligible	<i>Essentially no measurable repassivation within the test window — markedly worse than Ti-6Al-4V's partial +0.095 V recovery [6]</i>
WEAR-SBF ACCELERATION (TMZF ANALOGUE)	Significant, vs. dry wear	<i>Wear rate in SBF substantially exceeds dry-wear rate, unlike Ti-6Al-4V, where dry and wet wear track more closely [18]</i>
PASSIVE CURRENT TREND WITH FE CONTENT	Increases with Fe%	<i>Documented across the Ti-xFe alloy series; higher Fe content correlates with progressively poorer passive film performance [16]</i>

4.4 Where the Picture Is More Favourable

Two caveats are necessary for balance. First, **Ti-6Al-1V-3Fe** — an alloy with a more modest, balanced V/Fe content than Ti185's 8V-5Fe — performed comparably to (not worse than) Ti-6Al-4V in peak sliding current and exhibited a respectable +0.059 V anodic OCP recovery, indicating that V/Fe co-alloying is not invariably detrimental; the magnitude of V and Fe content appears to matter more than their mere presence.

[6] Second, the very high bulk hardness achievable in fully STA-aged Ti185 (up to 46–

50 HRC, see Section 4.4 of the main report) substantially raises the mechanical force threshold required to initiate oxide fracture in the first place — meaning that, in applications with bounded contact stresses below this higher threshold, the disrupted-oxide scenario may simply occur less often for Ti185 than for the softer Ti-6Al-4V baseline, even if its consequences are more severe once it does occur.

TI-8V / TI185 UNDER DISRUPTED-OXIDE SCENARIO – SUMMARY

Available analogue evidence indicates that once the oxide is broken, Fe/V-rich β -titanium alloys of this class show *weaker* depassivation resistance and repassivation capacity than the Ti-6Al-4V baseline, and the single-phase β microstructure introduces a wear-corrosion synergy (accelerated wear specifically in SBF, not seen in dry wear) that is mechanistically distinct from and additional to simple electrochemical dissolution. The higher V content (8% vs. 4%) compounds the concern by roughly doubling the dissolvable V_2O_5 reservoir in the passive film. This makes Ti-8V/Ti185 the less favourable of the two alloys specifically under the disrupted-oxide-in-SBF scenario, despite its much higher bulk hardness and mechanical strength — properties that govern resistance to *initiating* oxide breakage, not what happens once it occurs.

SECTION 5

Quantitative Comparison Tables

PARAMETER	TI-6AL-4V (BASELINE)	TI-6ZR	TI-8V / TI185
PASSIVE FILM COMPOSITION	TiO ₂ + Al ₂ O ₃ + V ₂ O ₅	TiO ₂ + ZrO ₂ (both stable)	TiO ₂ + 2× V ₂ O ₅ + Fe-oxide
RELATIVE OCP DROP (ΔE) ON SLIDING	Largest of comparator set (658 mV @ 17.5 N)	~Half of baseline (345–416 mV, Zr-richer analogue)	≥ Baseline (0.61 V, Fe-rich analogue)

PARAMETER	TI-6AL-4V (BASELINE)	TI-6ZR	TI-8V / TI185
REPASSIVATION CAPACITY	Partial (+0.095 V recovery)	Superior – explicitly reported as better than CP-Ti	Inferior — <0.01 V recovery in Fe-rich analogue
WEAR-CORROSION SYNERGY IN SBF	Moderate (α + β strain hardening helps)	Not directly studied; β -content low at 6% Zr	Significant — single-phase β lacks strain hardening (TMZF analogue)
TOXIC/BIOACTIVE IONS RELEASED	Al^{3+} , vanadate (VO_4^{3-})	None — Ti^{4+}, Zr^{4+} only	Vanadate (2 $\times$ baseline V content), Fe species
BULK HARDNESS (RAISES FRACTURE THRESHOLD)	~330–350 HV	240–280 HV (lower)	450–600+ HV (STA, much higher)
NET ASSESSMENT, DISRUPTED-OXIDE SCENARIO	Reference	More favourable than baseline	Less favourable than baseline

Table 1. Consolidated disrupted-oxide / tribocorrosion comparison. Ti-6Zr and Ti-8V data points are extrapolated from the closest available compositional analogues, as discussed in Sections 3–4; Ti-6Al-4V data is direct. Data compiled from refs. [3,5,6,7,13,14,18].

5.1 Relative OCP Collapse Magnitude (Illustrative Scale)

Static comparison of reported/extrapolated ΔE values on oxide disruption; lower bar = smaller potential collapse = more resilient passive film.



Mechanistic Synthesis — Why the Three Alloys Diverge

Three independent mechanisms explain the divergence observed once the oxide film is broken, and it is useful to separate them because they do not all point the same direction for every alloy:

(i) Oxide thermodynamic stability

ZrO₂ is a more stable oxide than TiO₂ itself, while Al₂O₃ and V₂O₅ are both prone to dissolution under physiological/mechanically disturbed conditions.^[7] This single factor favours Ti-6Zr and disfavours both Ti-6Al-4V and, more strongly (due to double the V content), Ti-8V/Ti185.

(ii) Microstructural strain-hardening response

Duplex $\alpha+\beta$ alloys (Ti-6Al-4V) and α -dominant alloys with limited β fraction (Ti-6Zr, which is essentially single-phase α at 6 wt% Zr) retain some capacity to work-harden at a freshly exposed sliding contact, partially resisting progressive material removal. Single-phase metastable β alloys (Ti-8V/Ti185, and by analogy TMZF) lack this response, producing a documented wear-acceleration specifically in SBF that is absent in dry wear testing of the same alloys.^[18] This factor disfavours Ti-8V/Ti185 specifically and is largely neutral for Ti-6Zr given its low overall β -phase content.

(iii) Toxicological profile of released species

Independent of corrosion rate, the *identity* of what dissolves matters clinically. Ti and Zr are both classed as biophile with no established adverse ionic pathway; Al and V are both linked in the literature to neurological and skeletal adverse effects on chronic exposure.^[9,10] This factor strongly favours Ti-6Zr, is the defining weakness of the Ti-6Al-4V baseline, and is compounded — not mitigated — in Ti-8V/Ti185 by the higher absolute vanadium content.

All three mechanisms point in the same direction for Ti-6Zr (consistently favourable relative to the baseline) but mechanisms (i) and (iii) disfavour Ti-8V/Ti185 while its much higher hardness (raising the force threshold for initiating oxide breakage in the first place) is the one factor working in its favour. This is why Section 4.4 above frames Ti-8V/Ti185's position as application-dependent: in low-contact-stress,

non-articulating structural applications it may rarely reach the disrupted-oxide regime at all; in any application involving cyclic micromotion or sliding contact at physiological loads, the evidence base points toward materially worse biocorrosion outcomes than either Ti-6Zr or the Ti-6Al-4V baseline once that regime is reached.

SECTION 7

Engineering Conclusions

For applications where mechanical disruption of the passive oxide is anticipated or unavoidable — modular implant junctions, fretting-prone fixation hardware, articulating or load-bearing biomedical contacts — the evidence assembled here favours Ti-6Zr over both the Ti-6Al-4V baseline and the Ti-8V/Ti185 family on every biocorrosion-relevant axis examined: smaller potential collapse on depassivation, faster and more complete repassivation, and release of exclusively non-toxic ionic species. This reinforces, from a different angle, the same conclusion already established for the static (intact-oxide) corrosion behaviour reported in the main document.

For Ti-8V/Ti185, the disrupted-oxide scenario surfaces a vulnerability that is not visible in the alloy's outstanding static mechanical properties: its single-phase metastable β microstructure, by closest available analogy with the well-documented TMZF case, appears prone to a wear-acceleration effect specifically in biological fluid that is not predicted by hardness or dry-wear testing alone, and its doubled vanadium content compounds the ion-release concern already present in the Ti-6Al-4V baseline. This is consistent with Ti185's actual commercial deployment (Section 7.2 of the main report) — predominantly in aerospace fasteners, springs, and structural components where biological fluid exposure is not a service condition — rather than in implantable biomedical devices.

EVIDENTIARY CAVEAT

No published study to date directly measures tribocorrosion of binary Ti-6Zr or Ti-8V/Ti185 in SBF under controlled, matched conditions against Ti-6Al-4V. The conclusions above are built from the closest compositional and microstructural analogues available in the peer-reviewed literature, combined with established oxide thermodynamics

and ion-toxicology data, and should be read as evidence-based engineering inference rather than as direct experimental confirmation for these two exact compositions. Where a specific design decision depends on this distinction, alloy-specific tribocorrosion testing in the intended service environment is recommended before final material selection.

REFERENCES

References

- [1] Mathew MT, Kerwell S, Lundberg HJ, Sukotjo C, Mercuri LG. "Tribocorrosion and oral and maxillofacial surgical devices." *British Journal of Oral and Maxillofacial Surgery*, and related general tribocorrosion methodology reviews cited across refs [13],[19]–[28] in: "Fretting corrosion behaviour of Ti–6Al–4V alloy in artificial saliva." *Wear* (2010). Defines passive-film fracture/removal as mechanically driven, repassivation as chemistry-driven.
- [2] Ryu JJ, Cudjoe E, Patel MV, Caputo M. "Sliding corrosion fatigue of metallic joint implants: A comparative study of CoCrMo and Ti6Al4V in simulated synovial environments." *Lubricants* 10(4):65 (2022). Cumulative dose effects of repeated mechanical disruption.
- [3] "Tribocorrosion behavior of additive manufactured Ti-6Al-4V biomedical alloy" / related Ti-29Nb-21Zr, Ti-15Zr, Ti-15Zr-5Mo fretting-corrosion study. *ResearchGate* (2017). Ti-15Zr and Ti-15Zr-5Mo show advantageous properties and better recovery capacity of the passive oxide layer vs. CP-Ti G4.
- [4] "The Tribocorrosion Behaviour of Ti-6Al-4V Alloy: The Role of Both Normal Force and Electrochemical Potential." *ResearchGate* (2022). Passive current $<1 \times 10^{-6}$ A; sliding current $\sim 100 \times$ passive current; depassivation/repassivation current transients.
- [5] "Enhanced Wear Resistance and Tribocorrosion Performance of New Ti–Zr–Nb–Ta and Ti–Zr–Nb–Mo Alloys for Biomedical Implants." *Journal of Bio- and Tribo-Corrosion*, Springer Nature (2026). $\Delta E = 658 \pm 64$ mV for Ti-6Al-4V at 17.5 N (highest of comparator set); Ti-30Zr-5Nb-14Mo $\Delta E = 345\text{--}416$ mV.
- [6] Rundora N, Klenam D, Falodun O, Mathabathe M, Sibanda V, Bodunrin M. "Tribocorrosion response of Fe-modified titanium alloys in simulated body fluids with varying glucose concentrations." *Journal of Applied Biomaterials & Functional Materials* (2026). Ti-3Fe, Ti-4.5Al-1V-3Fe, Ti-6Al-1V-3Fe vs. Ti-6Al-4V in HBSS: OCP recovery, peak sliding current, specific wear rate data.
- [7] "Passive Layers and Corrosion Resistance of Biomedical Ti-6Al-4V and β -Ti Alloys." *Materials (MDPI)* 11(5):487 (2021). $\text{Al}_2\text{O}_3/\text{V}_2\text{O}_5$ dissolution in Ti-6Al-4V passive film; $\text{ZrO}_2/\text{Nb}_2\text{O}_5/\text{Ta}_2\text{O}_5$ greater stability than TiO_2 ; β -Ti corrosion mechanisms.
- [8] "Corrosion behavior of plasma electrolytically oxidized gamma titanium aluminide alloy in simulated body fluid." *PMC* (2016). Vanadium oxide dissolution mechanism on Ti-6Al-4V surface; chloride-accelerated defect formation.
- [9] "Electrochemical Corrosion Behavior of Ta₂N Nanoceramic Coating in Simulated Body Fluid." *PMC5457118*. Metal ion release effects: peripheral neuropathy, osteomalacia, Alzheimer's disease associations; wear debris and aseptic loosening from Ti-6Al-4V.
- [10] "Corrosion Passivation in Simulated Body Fluid of Ti-Zr-Ta-xSn Alloys as Biomedical Materials." *PMC10342382*. Al^{3+} and V^{5+} released as free radicals; clinical rationale for replacing Al/V with non-toxic alternatives.

- [11] "Corrosion Passivation in Simulated Body Fluid of Ti-Zr-Ta-xSn Alloys as Biomedical Materials." *PMCI0342382* (2023). Passivation improves with stabilising alloying additions to Ti-Zr base; chronoamperometric and impedance data.
- [12] "Influence of Tantalum Addition on the Corrosion Passivation of Titanium-Zirconium Alloy in Simulated Body Fluid." *PMCI0342382* (2022). Ti-15Zr base alloy with Ta addition; SBF exposure 1 h and 72 h; cyclic polarisation, impedance.
- [13] "Tribocorrosion properties of different type titanium alloys in simulated body fluid." *Wear / ScienceDirect* (2014). Ti-13Nb-13Zr (β) and Ti-6Al-4V (α/β) showed lower volume loss and lower friction coefficient than other alloys tested (incl. β -Ti15Mo, β -Ti45Nb) under OCP/cathodic/anodic potentials.
- [14] Grandin HM, Berner S, Dard M. "A Review of Titanium Zirconium (TiZr) Alloys for Use in Endosseous Dental Implants." *Materials* 5:1348–1360 (2012). Ti and Zr biophile classification; in-vitro cytotoxicity testing summary across Ti-Zr compositions.
- [15] Okazaki Y, Gotoh E. "Comparison of metal release from various metallic biomaterials in vitro." *Biomaterials* 26:11–21 (2005). Cell viability >90% for Ti-xZr extraction media; absence of established Zr neurotoxic pathway.
- [16] "Exploring the role of Fe in the wear and corrosion resistance of Ti-based alloys produced with conventional powder metallurgy." *ScienceDirect* (2024). Passive current density increases with Fe content in Ti-xFe series; comparable passivation of Ti-6Al-4Fe/Ti-5Al-2.5Fe to Ti-6Al-4V at lower Fe loadings.
- [17] "Vanadium Oxide – an overview." *ScienceDirect Topics*; "vanadium oxide compounds: Topics by Science.gov." V₂O₅ classified as a Lux-Flood acidic oxide; oxidation states +II to +V; comparative chemical stability data vs. other ceramic oxides including ZrO₂.
- [18] "Predominant factor determining wear properties of β -type and ($\alpha+\beta$)-type titanium alloys in metal-to-metal contact for biomedical applications." *ScienceDirect* (2014). TMZF (Ti-12Mo-6Zr-2Fe) vs. Ti-6Al-4V: identical dry-wear behaviour but significantly accelerated wear in SBF for TMZF; lack of strain-hardening capacity in single-phase β -Ti identified as the mechanism; direct context of FDA recall investigation.